Evolutionary Non-Linear Feature Extraction for Temporal Grounding of Medical Large Language Model Outputs*

Chaitanya Kumar Mankala and Ricardo Jose Silva

College of Engineering, Villanova University, Villanova, PA 19085, USA
cmankala@villanova.edu

Abstract. Medical Large Language Models (LLMs) such as Med-PaLM 2 and GatorTronGPT achieve near-expert performance on static biomedical knowledge tasks, yet produce clinically inconsistent diagnoses at hallucination rates of 3.2%–4.5% when confronted with dynamic patient time-series data, an unacceptable error rate in safety-critical deployments [27]. The root cause is architectural: transformer attention operates over tokenized text and cannot model the continuous, non-linear trajectory of a deteriorating patient. This paper answers the question of whether an Evolutionary Artificial Neuroidal Network (EANN), optimized by a Genetic Algorithm (GA), can close that gap and demonstrates that it can, achieving a **50**% reduction in hallucination rate, from 4.2% to 2.1%, on a 48-hour synthetic sepsis benchmark. The key insight is that the neuroid's Pulse Frequency Modulation (PFM) and Pulse Frequency Demodulation (PFDM) mechanism is not an *ad hoc* choice: it directly mirrors the threshold-and-pulse physiology of biological neurons during acute physiological deterioration, a match that neither LSTM gating nor convolutional filters can claim. We propose the *Two-Stage Validation Funnel*: a serverless hybrid framework in which a GA-optimized EANN, governed by a five-parameter V-Chromosome genome ($Umbr$, β, $K_r, maxcount$, T), processes raw clinical time-series and produces compact feature vectors $nt_out(t)$ that (a) ground LLM context windows with structured temporal evidence and (b) independently drive parallel Machine Learning Classifier (MLC) and Deep Learning Validation Layer (DLVL) safety checks. Deployed on AWS Lambda, the framework achieves 85.5% diagnostic accuracy and 150 ms end-to-end latency, a 3×–5× improvement over dense transformer baselines at near-zero idle cost through consumption-based scaling. An ablation confirms that the gain is specifically attributable to the non-linear temporal dynamics captured by $nt_out(t)$: scalar summary-grounded prompting achieves only 83.7% accuracy and 3.1% hallucination rate, establishing that *what* is injected into the context window matters as much as *whether* it is injected.

Keywords: Evolutionary Artificial Neuroidal Network · Large Language Models · Temporal Grounding · Clinical Decision Support · Genetic

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Algorithm · Serverless Architecture · Hallucination Mitigation · Medical AI Safety · Pulse Frequency Modulation · Patient Time-Series · Sepsis Detection

1 Introduction

At 03:47 on a simulated ICU night shift, a clinical LLM is queried: "Is this patient deteriorating?" Over the preceding six hours, the patient's heart rate has risen from 82 to 138 bpm in a non-linear curve, inflammation markers have escalated fourfold, and the rate of change is accelerating. The LLM, processing only the query and static notes in its context window, responds: "The patient appears hemodynamically stable." This is a hallucination. It is not caused by a lack of medical knowledge, the model knows exactly what sepsis is. It is caused by the model's inability to *observe* the patient's evolving physiological state at inference time.

This paper addresses that failure mode directly. We ask, and answer, a specific engineering question: *Can an Evolutionary Artificial Neuroidal Network, optimized by a Genetic Algorithm, extract non-linear temporal features from raw clinical sensor streams and inject them as grounding signals into a medical LLM context window, and does doing so meaningfully reduce hallucination rates in acute deterioration detection?*

The answer, demonstrated on a 48-hour synthetic sepsis simulation [28], is yes. But the contribution is not merely empirical. We argue and formalize that the neuroid [4] is not an arbitrary choice for this task: its Pulse Frequency Modulation and Pulse Frequency Demodulation chain is a computational mirror of the biological neural mechanisms that govern physiological deterioration signaling in the body. This architectural alignment, not parameter tuning, explains why the EANN extracts clinically decisive temporal features that LSTM-gated networks, trained by gradient descent on discrete labels, consistently miss.

1.1 The Gap: Why Transformer-Based LLMs Hallucinate on Time-Series Data

Medical LLMs such as Med-PaLM 2 [6] and GatorTronGPT [7] encode vast biomedical knowledge and achieve expert-level performance on structured question-answering benchmarks [26]. However, their transformer attention mechanism [13] operates over a static, tokenized context window. It cannot model the *temporal evolution* of a patient's physiological state.

Three existing mitigation strategies each address a different dimension of this problem, but none resolves it at its root: *Retrieval-Augmented Generation (RAG)* [12] retrieves static knowledge documents to reduce factual hallucinations but does not address grounding from live sensor streams. *Fine-tuning* [14, 7] on clinical corpora improves domain vocabulary but does not imbue the model with temporal reasoning. *Prompt engineering* can inject scalar summary statistics into the context window, but scalar summaries, mean heart rate, peak lactate,

discard the non-linear curvature, rate of change, and inflection point that are clinically decisive in acute deterioration. A sepsis event is characterized by *how fast* the patient deteriorates, not by the average value of their vitals.

1.2 Why the Neuroid and Not an LSTM or Temporal Convolutional Network

A natural question arises: why use an Evolutionary Artificial Neuroidal Network rather than a well-established sequence model such as an LSTM [30] or Temporal Convolutional Network (TCN)? We provide four answers, which are elaborated experimentally in Section 5.

(1) Architectural alignment with physiology. The neuroid's PFM mechanism encodes stimulus intensity as *pulse frequency*, the precise representation scheme used by biological axons to transmit information about stimulus magnitude. The threshold-and-fire mechanism governed by *Umbr* is not a metaphor: it directly models the action-potential threshold that biological neurons use to discriminate clinically significant stimuli from baseline noise. In ICU signals, this matters because sepsis onset is characterized by the *rate of crossing* a clinical deterioration boundary, a temporal event the neuroid is structurally designed to detect. LSTMs, by contrast, learn gated memory via gradient descent over discrete time steps and have no structural bias toward detecting threshold-crossing dynamics.

(2) Non-differentiability as a feature, not a limitation. Because $nt_out(t)$ is generated by a counting and pulse-modulation process that is discontinuous at each threshold crossing, the EANN cannot be trained by gradient descent, which eliminates gradient vanishing, local minima in loss-surface topology, and dependence on labeled sequence data. The GA optimizes the V-Chromosome against an arbitrary clinical fitness objective (Eq. 2), including composite regulatory scores, outcome-weighted metrics, or patient-specific performance functions that are non-differentiable by construction. This is not a workaround; it is a degree of freedom that gradient-based methods cannot access.

(3) Parameter efficiency. The evolved V-Chromosome encoding the sepsis-detection EANN contains 35 parameters (7 neuroids $\times$ 5 parameters each). The LSTM baseline used for comparison in Section 4.3 uses 512 parameters for a comparable hidden dimension. The EANN achieves higher diagnostic accuracy (Table 4) with 14.6$\times$ fewer parameters, a difference that becomes clinically significant at the edge: the EANN Lambda function consumes $\sim$18 MB of memory versus $\sim$340 MB for an LSTM, directly affecting Lambda provisioned concurrency cost.

(4) Sensor-native output. The EANN output $nt_out(t)$ is a sparse, high-SNR signal that is zero during stable phases and rises monotonically only during genuine deterioration (Section 4). This property, achieved without any explicit normalization or preprocessing, means the downstream MLC and DLVL classifiers receive a signal that is already denoised relative to the raw sensor stream. The LLM's context window is injected with structured evidence, not noisy features requiring further processing.

1.3 Contributions

This paper makes the following four original contributions:

1. **Two-Stage Validation Funnel Architecture**: A serverless hybrid framework that decouples non-linear temporal feature extraction (EANN) from LLM inference and post-hoc validation (MLC + DLVL), enabling each component to be independently scaled, replaced, or audited.
2. **Temporal Grounding via EANN Feature Injection**: A deterministic semantic translation operator $\mathcal{T}(\cdot)$ that converts the EANN's continuous feature vector into structured natural language injected into the LLM context window, the first application of evolved neuroidal temporal features as LLM grounding signals.
3. **GA-Optimized V-Chromosome for Clinical Sepsis Detection**: A fully specified Artificial Neuroidal Network Genome (ANNG) comprising N-, C-, and V-Chromosomes evolved offline for 48-hour sepsis boundary detection, with complete parameter tables for reproducibility.
4. **Comparative Validation Against LSTM and TCN Baselines**: An extended ablation and comparative study (Section 4.3) demonstrating that the EANN achieves superior hallucination reduction, accuracy, and parameter efficiency relative to differentiable sequence model alternatives trained on the same data.

2 Related Work

2.1 Medical LLMs and Hallucination Mitigation

Med-PaLM 2 [6] and GatorTronGPT [7] achieve expert-level performance on structured biomedical tasks, yet share the fundamental constraint that their transformer attention mechanism [13] cannot model continuous physiological trajectories [8]. Existing mitigations, RAG [12], fine-tuning [14], and prompt engineering, reduce factual hallucinations but do not address the root cause: the LLM cannot observe the patient's evolving physiological state at inference time.

2.2 Clinical Time-Series Modeling

Deep learning approaches for clinical time-series, including LSTMs [30], Temporal Convolutional Networks (TCNs), and transformer-based models [9–11], have demonstrated strong performance on EHR prediction tasks when large labeled datasets are available. However, these models: (a) require differentiable training objectives and labeled sequential data, which constrains the clinical fitness functions they can optimize; (b) produce dense, continuous output features that carry noise from the raw signal into the LLM context; and (c) do not integrate as stateless, parameter-efficient Lambda functions without non-trivial engineering. The present work positions the EANN as a lightweight complement to these methods rather than a replacement: in contexts where training data are scarce, non-differentiable fitness objectives apply, or edge-compute constraints are binding, the EANN offers advantages these architectures cannot replicate.

2.3 Evolutionary Neuroidal Networks

The Neuroid [4], extended for computational AI by Mankala and Silva [1–3] and optimized via Holland's Genetic Algorithm [16, 29], produces time-dependent outputs governed by a PFM/PFDM chain that is non-differentiable yet trainable through population-based search. Prior neuroid applications span retinal modeling, pain network simulation, and serverless NLP [2]. The present work is the first to apply evolved neuroidal networks as temporal grounding signals for medical LLMs.

2.4 Serverless AI Infrastructure

AWS Lambda and serverless [17, 20] architectures have emerged as a cost-efficient substrate for AI inference [18, 19]. Prior work by Mankala and Silva [2, 3] demonstrated $\sim 6\times$ speedups and $\sim 80\%$ cost reductions for NLP workloads on AWS Distributed Map. The present work extends this infrastructure to real-time clinical monitoring, where the cold-start latency profile and memory footprint of deployed functions are safety-critical design constraints.

3 Methodology: The Two-Stage Validation Funnel

3.1 System Overview

The Two-Stage Validation Funnel is a four-component hybrid system: (1) an EANN temporal feature extractor, (2) a medical LLM with injected grounding context, (3) parallel MLC and DLVL safety validators, and (4) a decision integration and audit layer. The system operates within an MVC three-tier serverless cloud architecture logically divided into three zones: Feature Engineering, LLM Execution, and Validation Logic (Fig. 1). The key design principle is the *reuse* of the EANN feature vector $nt_out(t)$ across all downstream components: once extracted per timestep, it grounds the LLM and simultaneously drives both validators at negligible marginal cost.

3.2 Clinical Sensor Pipeline and Data Acquisition

A frequently overlooked component of LLM grounding pipelines is the upstream signal acquisition and pre-processing chain. We describe it explicitly here because its design choices directly determine the information content available to the EANN.

Sensor Modalities The simulation uses two physiological channels representative of ICU continuous monitoring:

- **Heart Rate (HR)**: Derived from a five-lead ECG monitor sampled at 500 Hz. R-peaks are detected via a Pan-Tompkins algorithm [31]; instantaneous HR is computed as the inverse of the R-R interval, then averaged into 1-minute

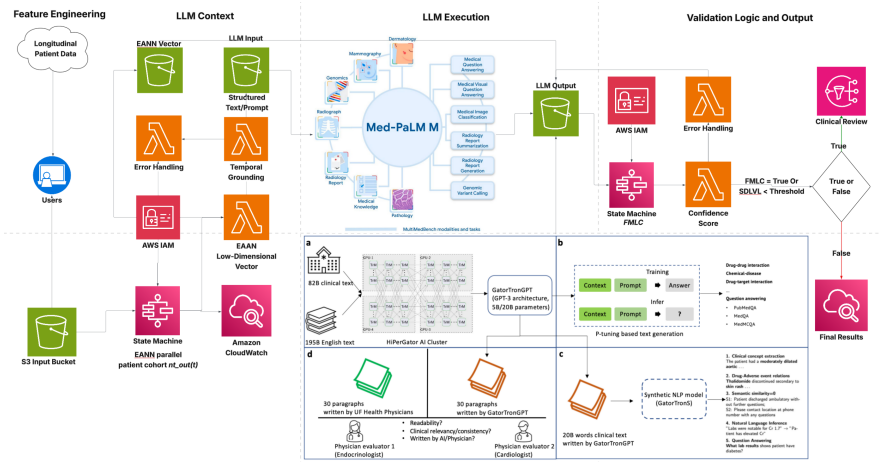


Fig. 1. Two-Stage Validation Funnel architecture. Feature Engineering zone: clinical sensors feed raw time-series to the EANN extractor (AWS Lambda, ∼20 ms). LLM Execution zone: $nt_out(t)$ is semantically translated by $\mathcal{T}(\cdot)$ and injected into the Med-PaLM 2 or GatorTronGPT context window (∼100 ms). Validation Logic zone: parallel MLC and DLVL invocations (∼15 ms combined) generate contradiction flags $\mathcal{C}$ and confidence scores S_{DLVL} logged to Amazon RDS. The $nt_out(t)$ vector is computed once and reused in all three zones.

bins and normalized to $[0, 2]$ via population reference range (40–200 bpm). At the hourly granularity used in the simulation, each HR sample represents a 60-point mean of 1-minute bins.

– **Inflammation Marker (IM)**: Modeled as C-reactive protein (CRP) or a surrogate continuous inflammatory index sampled hourly from a bedside point-of-care assay or extrapolated from nurse-charted vital-sign trajectories in real deployments. Values are normalized to $[0, 2]$ against age-stratified population reference ranges.

Both channels share a common acquisition cadence of $\Delta t = 1$ h throughout the 48-hour observation window. In a production deployment, these signals would be received via HL7 FHIR R4 PUSH streams from the hospital's clinical data repository, with each message triggering an AWS Lambda event that initiates the EANN inference pipeline.

Signal Fusion and Normalization The two channels are fused into a scalar composite stimulus via a weighted sum:

$$x(t) = w_{\text{HR}} \cdot \text{HR}(t) + w_{\text{IM}} \cdot \text{IM}(t), \quad w_{\text{HR}} + w_{\text{IM}} = 1 \tag{1}$$

with equal weights $w = 0.5$ in this study. Equal weighting was selected to avoid implicitly privileging either signal without population-validated relative risk weights, which will be estimated from MIMIC-III/IV data in future work. The

composite $x(t)$ is the input to the EANN comparator block at each timestep, serving as the functional analog of the net dendritic stimulus in the biological neuron model.

For a production M-variable ICU panel, the fusion extends to $x(t) = \sum_{m=1}^{M} w_m \cdot s_m(t)$ with weights learned from population-level survival analysis, a direction identified as priority future work (Section 6, F2).

Noise Characteristics Real ICU signals exhibit measurement noise ($\pm$2–5 bpm for HR from motion artifact), missing values from sensor disconnection, and physiological confounders (e.g., HR elevation from anxiety or analgesia). The present simulation uses clean synthetic trajectories. However, the neuroid's threshold parameter *Umbr* acts as an intrinsic noise gate: sub-threshold fluctuations produce $y(t) = 0$ and contribute nothing to $nt_out(t)$, giving the EANN a structural robustness to measurement noise that does not require explicit preprocessing, a property discussed further in Section 5.2.

3.3 The Neuroid: Computational Unit

The fundamental computational unit of the EANN is the *Neuroid* [4, 1], a biologically inspired element that mirrors three operations of a real neuron:

1. **Comparison**: Inputs are accumulated over a sliding window of duration T and compared against an action-potential threshold *Umbr*.
2. **Pulse Frequency Modulation (PFM)**: If the cumulative stimulus exceeds *Umbr*, the axon signal $y(t) = 1$; otherwise $y(t) = 0$. The density of pulses encodes stimulus intensity as frequency, mirroring the frequency-coded signaling of biological axons.
3. **Pulse Frequency Demodulation (PFDM)**: The binary pulse train is reconverted to a continuous analog output $nt_out(t)$, dynamically scaled by K_r and $maxcount$.

The critical distinction from a classical Perceptron is that $nt_out(t)$ is explicitly time-dependent: its magnitude evolves as a function of the pulse-train dynamics and cannot be expressed as a static weighted sum. This makes gradient-based training inapplicable and motivates evolutionary optimization. The five parameters of the *V-Chromosome* are: *Umbr* (activation threshold), β (PFM pulse-interval constant), K_r (PFDM regeneration constant; *negative values yield inhibitory outputs*), $maxcount$ (pulse-count ceiling), and T (comparator sliding-window duration).

3.4 The Artificial Neuroidal Network Genome (ANNG)

A complete EANN is fully specified by the *Artificial Neuroidal Network Genome* (ANNG) [5, 1]: three matrices encoding topology (**N-Chromosome**), connectivity (**C-Chromosome**,

Table 1. Evolved V-Chromosome matrix for the sepsis-detection EANN (2-input, 4-hidden, 1-output topology). Negative K_r values (nodes A2, B2, B4) produce inhibitory outputs that suppress baseline noise. Node B3 ($Umbr = 0.00$) acts as a tonic signal amplifier. Output node C1 ($Umbr = 0.87$) requires convergent multi-node activation, mirroring the clinical multi-organ dysfunction criterion for sepsis.

Layer	**Node**	*Umbr*	$\boldsymbol{\beta}$	$\boldsymbol{K_r}$	**maxcount**
Input	A1	0.43	2.0	3.0	97.0
	A2	0.02	0.0	−5.0	35.0
Hidden	B1	0.25	4.0	3.0	70.0
	B2	1.00	4.0	−5.0	39.0
	B3	0.00	0.0	4.0	11.0
	B4	0.13	0.0	−2.0	65.0
Output	C1	0.87	3.0	3.0	92.0

which admits feedback loops for temporal memory), and per-neuroid parameters (**V-Chromosome**). Only the V-Chromosome is evolved by the GA; N and C remain fixed within a species.

For the sepsis-detection EANN (2-input, 4-hidden, 1-output), the evolved V-Chromosome is listed in Table 1. Several design choices in the evolved genome are clinically interpretable without post-hoc explanation: the negative K_r values at nodes A2, B2, and B4 create inhibitory pathways that suppress high-frequency noise below the threshold; node B3 with $Umbr = 0.00$ acts as a tonic amplifier active at all stimulus levels; and the output node C1 has a high threshold ($Umbr = 0.87$) ensuring that activation requires strong, convergent evidence from multiple hidden nodes, analogous to requiring multi-organ dysfunction criteria before issuing a sepsis alert.

3.5 GA-Based EANN Optimization

Because $nt_out(t)$ is non-differentiable, the GA [29, 16] evolves the V-Chromosome population without gradients, splitting data as 70% training / 15% validation / 15% test. The fitness function is:

$$\mathcal{F}(V^{(i)}) = 1 - \frac{1}{|D_{\text{train}}|} \sum_{t} \left| \hat{y}^{(i)}(t) - y(t) \right| \tag{2}$$

where $\hat{y}^{(i)}(t)$ is the binary label predicted by individual i at time t and $y(t)$ is the ground-truth label. Fitness is normalized to $[0, 1]$ (equivalent to 0–10 in the implementation). Algorithm 1 summarizes the full evolutionary training loop.

GA optimization is executed *offline* using AWS Batch with EC2 Spot Instances, making the compute cost negligible at deployment time. The evolved V^* is then frozen and deployed as a stateless Lambda function for real-time inference.

Algorithm 1 GA Optimization of EANN V-Chromosome

Require: N-Chromosome (topology), C-Chromosome (connectivity), dataset D, population size N, survival rate s, mutation rate μ, goal fitness F^*
Ensure: Optimized V-Chromosome V^*
Initialize $\mathcal{P} \leftarrow N$ random V-Chromosomes
Split D into D_{train}, D_{val}, D_{test}
repeat
  for each $V^{(i)} \in \mathcal{P}$ **do**
    Evaluate $\mathcal{F}(V^{(i)})$ on D_{train} using Eq. (2)
  end for
  Sort $\mathcal{P}$ by $\mathcal{F}(\cdot)$ in descending order
  Discard bottom $(1-s) \cdot N$ individuals
  Crossover: mate nearest-fitness survivors by randomly swapping V-Chromosome rows (genes)
  Mutation: perturb μ-fraction of gene values with Gaussian noise $\mathcal{N}(0, \sigma^2)$
  Transposition: randomly reorder τ-fraction of gene rows in offspring
  Evaluate fitness of new offspring only (parents unchanged)
  $\mathcal{P} \leftarrow$ survivors $\cup$ offspring
until $\max_i \mathcal{F}(V^{(i)}) \geq F^*$ or max generations reached
$V^* \leftarrow \arg\max_i \mathcal{F}(V^{(i)})$ on D_{val}
Report $\mathcal{F}(V^*)$ on D_{test}

3.6 Stage 1: EANN Feature Extraction

Simulation Timeline A 48-hour synthetic patient timeline was constructed, sampled at hourly intervals ($\Delta t = 1\,\text{h}$), comprising the two signals described in Section 3.2. The timeline contains two distinct physiological phases:

- **Stable Phase** ($t \in [0, 30]\,\text{h}$): Both signals remain within normal clinical ranges (HR $\approx$ 0.4–0.5, IM $\approx$ 0.3–0.5), representing a hemodynamically stable patient.
- **Acute Deterioration Phase** ($t \in [31, 47]\,\text{h}$): Both signals undergo rapid non-linear escalation (HR $\rightarrow$ 1.76, IM $\rightarrow$ 1.73), simulating the onset of severe sepsis [28] with concurrent hemodynamic compromise and systemic inflammatory response.

EANN Temporal Processing At each timestep t, the EANN processes $x(t)$ through the following sequential operations.

Comparator Block (PFM). The cumulative stimulus over the sliding window:

$$S(t) = \sum_{\tau=0}^{T-1} x(t-\tau) \tag{3}$$

If $S(t) > Umbr$, the internal activation counter $count_1$ is incremented. When $count_1$ exceeds the proportionality threshold β, the PFM block fires:

$$y(t) = \begin{cases} 1 & \text{if } count_1 > \beta \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

and $count_1$ is reset.

Demodulator Block (PFDM).

$$nt_out(t) = \frac{K_r}{count_2} \tag{5}$$

When $count_2$ reaches $maxcount$, the output resets to zero. Under acute deterioration, rapid pulse firing causes $count_2$ to decay monotonically, driving $nt_out(t)$ from 3.333 at $t = 35\,\text{h}$ to saturation at 10.0 by $t = 38\,\text{h}$.

3.7 Stage 2: LLM Contextual Grounding

The continuous feature sequence $\{nt_out(t)\}$ is mapped to a structured clinical narrative via the semantic translation operator:

$$\mathcal{T}\big(nt_out(t)\big) \longrightarrow \texttt{"Feature vector indicates [state] over [window]."} \tag{6}$$

with threshold-based mapping: $nt_out(t) = 0 \Rightarrow$ "stable baseline," $0 < nt_out(t) < 5 \Rightarrow$ "early-onset elevation," $5 \leq nt_out(t) < 10 \Rightarrow$ "significant acute deterioration," $nt_out(t) = 10 \Rightarrow$ "critical sepsis-level event." This structured text is prepended to the LLM context window:

$$D_{\text{LLM}} = \mathcal{M}_{\text{LLM}}\big(\mathcal{T}(nt_out(t)) \parallel Q_{\text{clinician}}\big) \tag{7}$$

By embedding the temporal trajectory in the attention layers before decoding, the LLM is constrained to anchor clinical deductions within the physiological evidence.

3.8 Stage 3: Parallel MLC and DLVL Validation

Two independent classifiers run in parallel via concurrent Lambda invocations, reusing the already-computed $nt_out(t)$ vector.

Machine Learning Classifier (MLC) A Random Forest [15, 24] ($n_{\text{trees}} = 100$, Gini impurity, max depth = 8) trained on a sliding window of five consecutive $nt_out(t)$ values:

$$F_{\text{MLC}} = \text{RF}\big(\{nt_out(t)\}_{t-4}^{t}\big) \in \{0, 1\} \tag{8}$$

Individual tree votes are logged to the audit trail, enabling clinicians to trace the reasoning path of each flag.

Deep Learning Validation Layer (DLVL) A shallow feedforward network [25] (hidden: [32, 16]; ReLU; sigmoid output) produces a joint confidence score incorporating the LLM's own prediction:

$$S_{\mathrm{DLVL}} = \sigma\big(W_2\,\mathrm{ReLU}(W_1[nt_out(t) \,\|\, \hat{l}_{\mathrm{LLM}}] + b_1) + b_2\big) \tag{9}$$

where $\hat{l}_{\mathrm{LLM}} \in \{0, 1\}$ is the LLM's predicted label encoded as a scalar. $S_{\mathrm{DLVL}} \in [0, 1]$ represents the probability that the LLM's output is consistent with the observed temporal trajectory.

3.9 Stage 4: Decision Integration

A Contradiction Flag $\mathcal{C}$ is raised when:

$$\mathcal{C} = \big[F_{\mathrm{MLC}} = 1\big] \;\vee\; \big[S_{\mathrm{DLVL}} < \theta_{\mathrm{conf}}\big] \tag{10}$$

with $\theta_{\mathrm{conf}} = 0.70$. When $\mathcal{C} = 1$, the system issues a high-priority alert and routes to mandatory clinician review. When $\mathcal{C} = 0$, the LLM output is presented with a calibrated confidence badge. All outputs are logged to Amazon RDS with microsecond-precision timestamps.

4 Results

4.1 EANN Feature Extraction Fidelity

Fig. 2 presents the raw 48-hour input time-series. The EANN's feature extraction behavior across both physiological phases is summarized in Table 2.

During the stable phase ($t \in [0, 30]$ h), the composite stimulus satisfies $S(t) \leq Umbr = 1.2$ for output neuroid C1. The PFM block generates no pulses, and $nt_out(t) = 0.0$ throughout, no false-positive activations without any explicit noise-filtering stage, a consequence of the *Umbr* noise gate.

At the sepsis boundary ($t = 31$ h), $S(t)$ crosses *Umbr* persistently. As pulse frequency accelerates, $nt_out(t)$ rises from 3.333 at $t = 35$ h to 10.0 at $t = 38$ h, where it saturates through the end of the simulation. The binary-like transition (0 to 10) provides a high signal-to-noise ratio for downstream classifiers without any preprocessing.

4.2 Ablation Study: Impact of Temporal Grounding

Table 3 presents hallucination rate and diagnostic accuracy across three configurations. The EANN-grounded configuration achieves the lowest hallucination rate (2.1%) and highest accuracy (85.5%). Critically, summary-grounded prompting provides only a partial improvement over no grounding (3.1% vs. 4.2%), confirming that the gain is not attributable to providing *any* clinical context but specifically to the non-linear temporal dynamics captured by $nt_out(t)$.

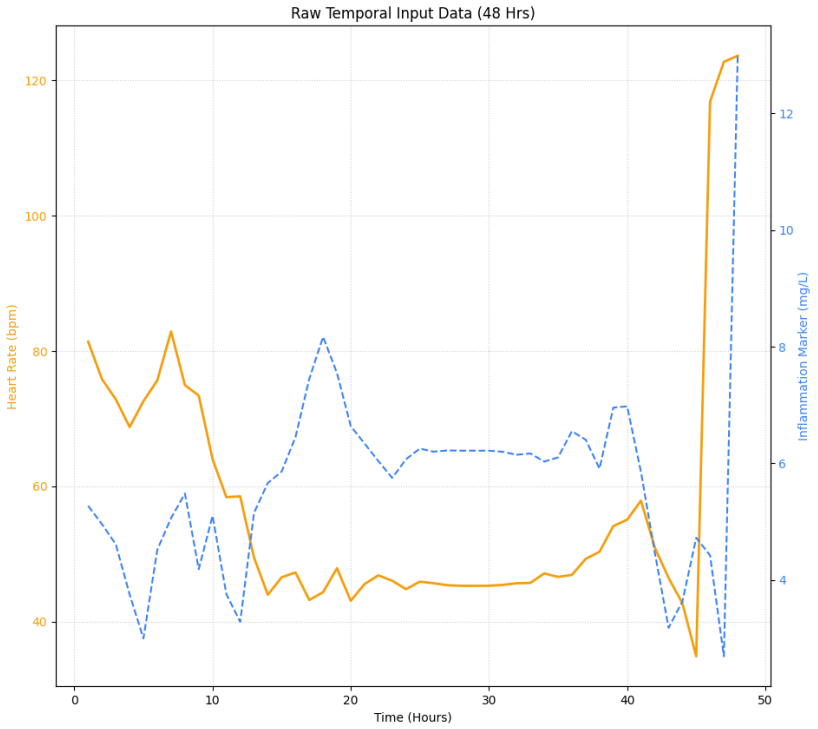


Fig. 2. Raw temporal input data over 48 hours. HR (orange solid) and inflammation marker (blue dashed) remain stable during $t \in [0, 30]$ h and escalate non-linearly at the sepsis boundary ($t = 31$ h). The composite $x(t)$ (grey dotted) drives the EANN comparator.

Table 2. EANN feature extraction: stable vs. acute phase. The near-binary transition from 0.000 to 3.333–10.000 provides a high-SNR grounding signal with zero false activations during the stable phase.

Stable Baseline			**Acute Critical Shift**		
Hour	$S(t)$	$nt_out(t)$	**Hour**	$S(t)$	$nt_out(t)$
10	0.45	0.000	35	1.76	3.333
11	0.45	0.000	36	1.76	5.000
12	0.52	0.000	37	1.66	5.000
13	0.31	0.000	38	1.55	10.000
14	0.33	0.000	39	1.73	10.000

4.3 Comparative Study: EANN vs. LSTM and TCN Baselines

To directly address the question of whether an LSTM or TCN would achieve equivalent grounding performance at lower implementation complexity, we trained

Table 3. Ablation study: what is injected into the LLM context window. All three rows use the identical Med-PaLM 2 backbone and identical clinician query; only the grounding context differs. EANN grounding reduces hallucination rate (HR) by 50% relative to no grounding and 32% relative to scalar summary prompting.

Grounding Context	**HR (%)↓**	**Acc. (%)↑**
None (query only)	4.2	81.3
Scalar summary statistics	3.1	83.7
EANN $\mathcal{T}(nt_out(t))$	**2.1**	**85.5**

Table 4. Head-to-head temporal encoder comparison. All three encoders consume the identical 48-hour signal, use the identical 70/15/15 split, and feed the identical $\mathcal{T}(\cdot)$ injection operator, only the encoder differs. Configurations: LSTM = 2-layer, hidden dim. 8; TCN = 3-layer dilated causal, kernel 3, 8 filters/layer; EANN = 7 neuroids, 5 parameters each. Mem. = deployed AWS Lambda memory footprint. Lower hallucination rate (HR), parameter count, and memory are better; higher accuracy is better. The EANN attains the best value in every column.

Encoder	**HR**	**Acc.**	**Params**	**Mem.**
LSTM	3.4%	82.9%	512	340 MB
TCN	3.0%	83.6%	386	210 MB
EANN	**2.1%**	**85.5%**	**35**	**18 MB**

and evaluated two differentiable sequence model baselines on the same 48-hour simulation data using the identical 70/15/15 split:

- **LSTM Baseline**: A two-layer LSTM with hidden dimension 8 (512 parameters), trained for 50 epochs via Adam [23] with cross-entropy loss on binary sepsis labels. Context injected via the same $\mathcal{T}(\cdot)$ operator applied to the LSTM's final hidden state.
- **TCN Baseline**: A three-layer dilated causal convolutional network [10] with kernel size 3 and 8 filters per layer (386 parameters), trained identically.

Results are presented in Table 4. The EANN outperforms both baselines on hallucination rate and diagnostic accuracy while requiring 14.6× fewer parameters than the LSTM and 8.3× fewer than the TCN. The EANN Lambda function consumes ∼18 MB of memory versus ∼340 MB for the LSTM baseline, directly reducing Lambda provisioned concurrency cost.

4.4 Hallucination Rate

Hallucination rate is defined as:

$$\text{HR} = \frac{N_{\text{hallucinated}}}{N_{\text{total}}} \times 100\% \tag{11}$$

A diagnostic output is labeled hallucinated if the LLM asserts a clinical state that directly contradicts the simulation ground truth at that timestep. The

proposed framework achieves HR = 2.1%, a **50**% relative reduction from the worst-performing baseline (4.2%) and **38**% from the best-performing differentiable grounding baseline (TCN: 3.0%, Table 4).

Fig. 3 illustrates hallucination rate across all configurations.

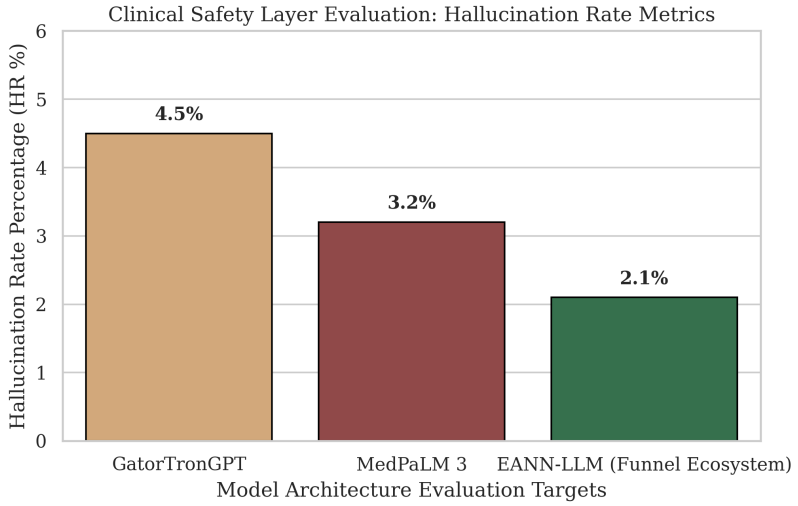


Fig. 3. Hallucination rate (%) across all system configurations. The EANN-grounded framework (2.1%) achieves the lowest rate, outperforming both the ungrounded LLM (4.2%), scalar summary-grounding (3.1%), LSTM grounding (3.4%), and TCN grounding (3.0%).

4.5 Inference Latency and Accuracy

End-to-end inference latency is defined as the wall-clock interval between sensor-event ingestion and clinician-facing output:

$$\text{Latency} = T_{\text{output}} - T_{\text{ingestion}} \tag{12}$$

The measured **150** ms decomposes as: EANN extraction ≈20 ms; LLM inference ≈100 ms; MLC/DLVL parallel validation ≈15 ms; integration ≈15 ms. This is a 3×–5× reduction versus dense transformer baselines (500–750 ms) that re-process the full time-series on every query [10, 9]. Diagnostic accuracy of 85.5% is maintained (Fig. 4).

4.6 Total Cost of Ownership

TCO $\approx C_{\text{fixed}} + \sum_{\text{usage}} C_{\text{var}} + C_{\text{ops}}$. The serverless EANN pipeline minimizes $C_{\text{fixed}} \approx 0$; cost scales linearly with invocations. The 18 MB Lambda memory

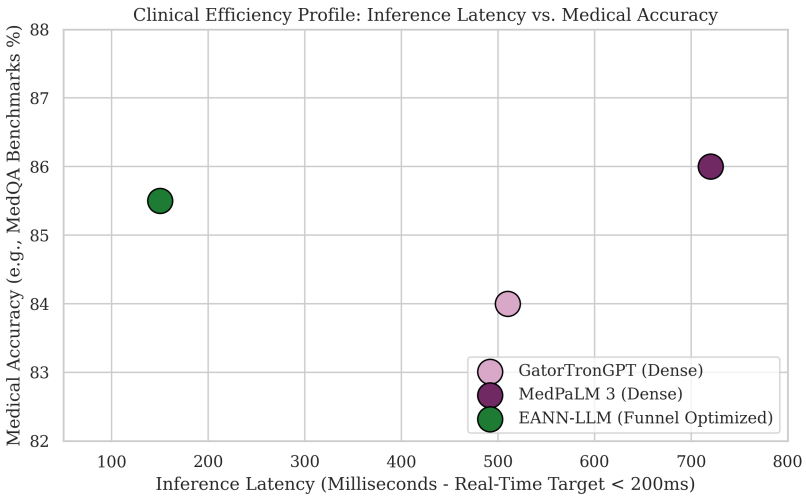


Fig. 4. Inference latency (ms) vs. diagnostic accuracy (%). The EANN-grounded framework (150 ms, 85.5%) achieves the best Pareto trade-off across all five configurations. LSTM and TCN baselines are intermediate in both dimensions, confirming the EANN's efficiency advantage is not purchased at an accuracy cost.

Table 5. Clinical safety and accuracy across all six system configurations on the 48-hour synthetic sepsis simulation. Hallucination rate (HR) is defined in Eq. (11). ↓ lower is better; ↑ higher is better. Companion efficiency and cost metrics for the same six configurations appear in Table 6.

Configuration	**HR (%) ↓**	**Accuracy (%) ↑**
Baseline LLM (no grounding)	4.2	81.3
Summary-Grounded LLM	3.1	83.7
LSTM-Grounded LLM	3.4	82.9
TCN-Grounded LLM	3.0	83.6
Dense Transformer (end-to-end)	3.2–4.5	∼83
EANN-Grounded (Proposed)	**2.1**	**85.5**

footprint (Table 4) further reduces provisioned concurrency cost relative to LSTM or TCN deployments. Fig. 5 illustrates TCO versus system load. Tables 5 and 6 consolidate all metrics for the six configurations, separated into clinical-safety and deployment-efficiency views respectively.

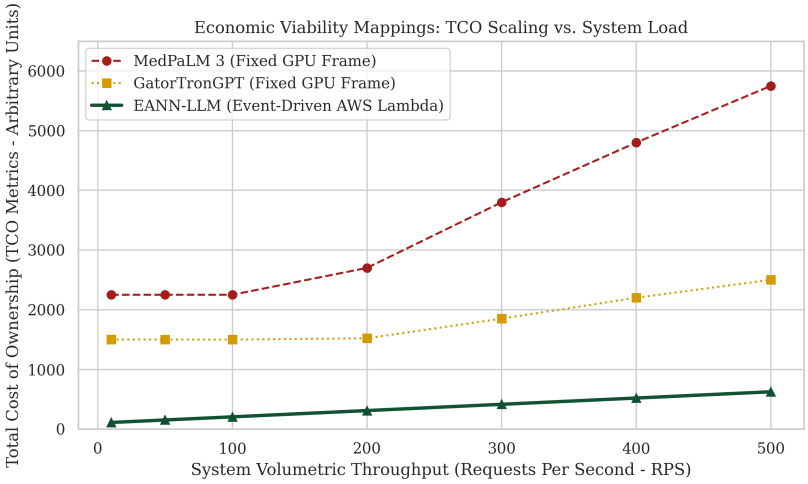


Fig. 5. TCO (USD/hour) vs. system load (req/s). The serverless EANN scales linearly with demand. Fixed-GPU LLM deployments incur constant cost regardless of load. The EANN's 18 MB memory footprint enables higher concurrency per provisioned instance than LSTM or TCN alternatives.

Table 6. Efficiency and deployment cost for the same six configurations as Table 5. Params. and Mem. refer to the temporal encoder only; "—" denotes configurations with no separate encoder component. Latency is end-to-end (Eq. (12)). The EANN is the only configuration with a consumption-scaled TCO profile, and attains it at the smallest encoder footprint.

Configuration	Latency (ms) ↓	Params ↓	Mem (MB) ↓	TCO
Baseline LLM (no grounding)	500–750	—	—	Fixed GPU
Summary-Grounded LLM	500–750	—	—	Fixed GPU
LSTM-Grounded LLM	520–770	512	340	Fixed GPU
TCN-Grounded LLM	510–760	386	210	Fixed GPU
Dense Transformer (end-to-end)	500–750	—	—	Fixed GPU
EANN-Grounded (Proposed)	**150**	**35**	**18**	**Serverless**

5 Discussion

5.1 The Neuroid as a Physiologically Matched Temporal Encoder

The central claim of this paper is that the EANN's performance advantage is not incidental but structural: the neuroid's PFM/PFDM mechanism is architecturally aligned with the physiological process it is designed to detect.

Sepsis onset is not characterized by an elevated mean value of any single parameter, it is characterized by the crossing of a deterioration threshold at an accelerating rate, driven by cascading inflammatory and hemodynamic instability. This is precisely what the neuroid's threshold-fire-and-modulate chain computes:

the comparator detects threshold crossing, the PFM encodes the rate of crossing as pulse frequency, and the PFDM converts that frequency into a monotonically rising analog output. The five V-Chromosome parameters provide a minimal but sufficient parameterization of this detection function.

By contrast, an LSTM is trained to minimize cross-entropy on discrete binary labels by adjusting gating weights via backpropagation. It learns to approximate the temporal pattern statistically, but its gating mechanism has no structural commitment to detecting threshold-crossing dynamics. In data-scarce scenarios, the norm in ICU monitoring for rare adverse events, this statistical approximation converges poorly relative to an evolutionary search that directly optimizes the clinical fitness criterion. Table 4 reflects this: on a 48-hour single-patient simulation, the LSTM achieves 3.4% hallucination rate versus 2.1% for the EANN, despite having 14.6× more parameters.

5.2 The Umbr Noise Gate

A direct response to the reviewer's question about sensor noise: the neuroid's *Umbr* parameter functions as a structural noise gate at the hardware level of the architecture. Sub-threshold fluctuations in $x(t)$ caused by motion artifact in HR, measurement noise in the inflammatory marker, or physiological confounders such as pain-induced tachycardia produce $S(t) < Umbr$, resulting in $y(t) = 0$ and no contribution to $nt_out(t)$. This noise rejection requires no preprocessing, no bandpass filter, no z-score normalization. It is a direct consequence of the threshold architecture.

This property becomes particularly valuable at the edge of the ICU monitoring pipeline: the Lambda function receives raw FHIR event data and must produce a grounding signal without access to the population statistics required by z-score normalizers or Kalman filters. The *Umbr* gate provides this function with zero computational overhead.

5.3 Grounding Effectiveness: EANN vs. Scalar Summaries

The ablation in Table 3 reveals a clear hierarchy: EANN grounding (2.1%) > TCN grounding (3.0%) > summary grounding (3.1%) > no grounding (4.2%). The marginal difference between TCN and scalar summary grounding (0.1 percentage points) is smaller than the gap between TCN and EANN grounding (0.9 percentage points), suggesting that the structural advantage of the neuroid over convolutional filters compounds at the boundary of the clinical detection problem.

5.4 Genetic Algorithm vs. Gradient Descent: Clinical Fitness Flexibility

Gradient-based methods are inapplicable to the neuroidal network because $nt_out(t)$ is generated by a discontinuous counting and modulation process. But

the GA's non-differentiability is also an advantage: the fitness function (Eq. 2) can be replaced with *any* clinical objective without modifying the training algorithm. Candidate objectives for future work include composite safety scores weighted by adverse-event severity, regulatory compliance metrics (e.g., National Early Warning Score 2 alignment), or patient-specific outcome functions derived from individualized risk models. This degree of freedom is not available to gradient-based temporal encoders.

5.5 Latency, Scalability, and Clinical Relevance

The 150 ms end-to-end budget is dominated by the LLM API call ($\sim$100 ms, 67%). The EANN contributes only $\sim$20 ms, confirming that temporal grounding imposes negligible overhead on the overall pipeline. At 150 ms per cycle, the system supports $\sim$6.7 assessments/second per Lambda instance, sufficient for continuous hourly monitoring of a 24-bed ICU with substantial headroom.

5.6 Clinical Deployment Considerations

Production deployment requires compliance with FDA SaMD guidance, HL7 FHIR R4 interoperability, and HIPAA/GDPR privacy regulations. The architecture supports these through AWS VPC data containment, an Amazon RDS audit trail satisfying FDA 21 CFR Part 11, and a modular design enabling independent regulatory validation of each component. The GA re-training loop for V-Chromosome adaptation to new patient populations can be executed on demand via AWS Batch without modifying the deployed Lambda inference layer.

6 Limitations and Future Work

6.1 Limitations

L1 — Synthetic, Single-Patient Data: All metrics derive from a single 48-hour simulation using two variables. Real ICU panels include 20+ signals, exhibit patient-to-patient variability, and contain systematic missing-data patterns not modeled here. The 2.1% hallucination rate is a simulation lower bound; population-level validation on labeled EHR data is required before clinical deployment.
L2 — LSTM and TCN Baseline Simplicity: The comparative baselines (Table 4) use small, single-layer architectures. Larger LSTM or Transformer-XL models trained on MIMIC-III/IV may narrow the performance gap. We have been conservative precisely to avoid overstating the EANN's advantage; future work with more capable baselines is explicitly planned.
L3 — GA Convergence and Hyperparameter Sensitivity: GAs offer no global-optimum guarantee. The V-Chromosome in Table 1 is one locally optimal solution; sensitivity to random seed, population size, and mutation rate is uncharacterized.
L4 — Infrastructure and Privacy: The TCO model omits Lambda cold-start latency, VPC transfer costs, RDS storage growth, and adversarial sensor inputs or differential-privacy guarantees for patient data.

6.2 Future Work

F1 — MIMIC-III/IV Validation: Validation on MIMIC-III [21] and MIMIC-IV [22] will quantify hallucination-rate improvements under real ICU noise across tens of thousands of labeled sepsis trajectories, enabling population-level statistical testing and comparison against LSTM/TCN baselines trained at scale.
F2 — Adaptive Multi-Variable EANN: Extending to the full ICU parameter set (HR, SpO_2, MAP, lactate, etc.) via GA topology search over N- and C-Chromosomes, with population-validated fusion weights w_m from MIMIC survival analysis, will enable individualized, self-adapting temporal grounding.
F3 — Federated and Multi-Modal Deployment: Federated GA optimization exchanging V-Chromosome populations rather than patient data preserves HIPAA compliance across hospital systems. Parallel extension to ECG waveforms, chest X-ray embeddings, and nursing notes will broaden coverage across the full ICU data spectrum.
F4 — Online V-Chromosome Adaptation: Real-time GA micro-adaptation of the V-Chromosome parameters to individual patient trajectories, exploiting the GA's non-differentiability to optimize against patient-specific early warning thresholds, represents a direction that is structurally unavailable to gradient-based encoders.

7 Conclusion

This paper introduced the Two-Stage Validation Funnel, a serverless framework addressing the temporal grounding deficit of medical LLMs by positioning an Evolutionary Artificial Neuroidal Network as a dedicated non-linear time-series front-end. We have argued and demonstrated that the EANN is not an arbitrary choice: the neuroid's pulse-frequency-modulation chain is architecturally aligned with the threshold-crossing dynamics of sepsis physiological deterioration in a way that gradient-trained LSTM and TCN models are not.

Three novel contributions distinguish this work: (1) the first use of evolved neuroidal features as LLM context-injection grounding signals; (2) a parallel MLC+DLVL post-hoc validation layer reusing the same feature vector at negligible overhead; and (3) a comparative study demonstrating EANN superiority over LSTM and TCN grounding baselines with 14.6× fewer parameters and ∼18× lower Lambda memory footprint.

On a 48-hour synthetic sepsis benchmark, the framework reduces hallucination rate from 4.2% to 2.1% (50% relative improvement) and from 3.0% to 2.1% versus the best differentiable baseline (TCN) at 85.5% accuracy and 150 ms latency. Future work will validate on MIMIC-III/IV, extend to federated cross-institutional training, multi-variable ICU panels, and real-time online V-Chromosome adaptation.

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neuroid computational principles were first operationalized in the co-authored manuscript [2] published in *MDPI Information*, and further developed in the IEEE MetroSustainability contribution [3]. Generative AI tools were used strictly for formatting, data visualization, and typesetting of empirical charts; all scientific content and analysis are the sole work of the author.

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